

CandidateSync Design Summary Author: Sourabi R Email: yourmail@example.com Date: 2026-06-30

Project Overview: CandidateSync ingests structured recruiter CSV data and unstructured resume content, normalizes values, merges into a canonical profile, adds confidence and provenance, validates output, and projects JSON based on runtime config.

Architecture: - CSV Parser - Resume Parser - Canonical Mapper - Merge Engine - Confidence Engine - Projection Layer - Output Validator - CLI + Logging + Reports

Merge Strategy: - CSV is the primary source for structured fields. - Resume enriches contact data and skills. - Conflicts are recorded in conflict_report.json for traceability.

Confidence: - Overall confidence is computed from merged profile heuristics. - Confidence is included when enabled by config.

Projection: - Config controls field selection, confidence, and provenance. - Minimal config hides metadata; default config includes full traceability.

Validation: - Required fields are enforced during output validation. - Missing required projected fields cause a validation error.

Edge Cases: - Missing resume file logs an error and continues safely. - Duplicate emails and phones are normalized and deduplicated. - Phone numbers are normalized to E.164.